

LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development

Experiment No.1

Course Code: 23CM4121

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1. TITLE

Design and Development of a College Information Portal Using HTML5 (Structure, Formatting, Tables, Forms, and Multimedia)

2. AIM

To design a static web page titled "College Information Portal" using HTML5, demonstrating text formatting, lists, hyperlinks, tables, images, forms, semantic elements, and multimedia (audio/video) embedding.

3. OBJECTIVE

- To create a valid, well-structured HTML5 document.
- To display college information using headings and paragraphs.
- To apply text formatting tags such as Bold, Italic, Underline, Superscript, and Subscript.
- To create ordered and unordered lists.
- To add hyperlinks to external websites.
- To design a student details table using table-related tags.
- To insert images with appropriate alternate text.
- To design a student registration form using various HTML5 input types.
- To use semantic elements such as header, nav, section, article, and footer.
- To embed audio and video content using HTML5 media elements.

4. THEORY

HTML5 (HyperText Markup Language, version 5) is the standard markup language used to create the structure and content of web pages. It introduces semantic elements that describe the meaning of content rather than just its appearance.

Text formatting tags such as `<b>`, `<i>`, `<u>`, `<sup>`, and `<sub>` are used to bold, italicise, underline, superscript, and subscript text respectively.

Lists are created using `<ol>` (ordered list) and `<ul>` (unordered list), with individual items defined using the `<li>` tag.

Tables are built using `<table>`, `<tr>`, `<th>`, and `<td>` tags to organise tabular data such as student records.

Forms are created using the `<form>` tag along with `<input>`, `<select>`, and `<textarea>` elements to collect user input. HTML5 introduces new input types such as email, date, number, color, and range, which provide built-in validation and native browser controls.

Semantic elements such as <header>, <nav>, <section>, <article>, and <footer> improve the readability and structure of the document and help browsers and search engines understand the page layout.

Multimedia elements <audio> and <video> allow embedding of sound and video content directly in the browser without third-party plugins, using the <source> tag to specify the media file and its type.

5. ALGORITHM / PROCEDURE

- 1 Start the HTML5 document with the <!DOCTYPE html> declaration.
- 2 Define the <head> section with the character set, viewport meta tag, and page title.
- 3 Create a <header> section containing the college banner image and page heading.
- 4 Add a <nav> section with links to different parts of the page.
- 5 Inside an <article>, add headings and paragraphs describing the college, applying text formatting tags (bold, italic, underline).
- 6 Create one ordered list for courses offered and one unordered list for facilities.
- 7 Add hyperlinks to the college website and to Google, opening in a new tab.
- 8 Create a student details <table> with columns Roll No, Name, Branch, and Email.
- 9 Insert the college logo image and a campus image, each with descriptive alt text.
- 10 Design a registration <form> with text, email, password, date, number, color, and range inputs, radio buttons for gender, checkboxes for hobbies, a dropdown for branch, and a textarea for comments.
- 11 Add Submit and Reset buttons to the form.
- 12 Embed an <audio> element and a <video> element, each with a <source> tag.
- 13 Add a <footer> with copyright information.
- 14 Save the file with a .html extension and open it in a web browser to verify the layout and functionality.
- 15 Test each section (formatting, lists, table, form, media) and verify the actual output against the expected output.

6. PROGRAM

college-portal.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>
<body>

  <!-- Header -->
  <header style="text-align:center;">
    

    <h1>College Information Portal</h1>
    <p>Welcome to the College Information Portal.</p>
  </header>

  <!-- Navigation -->
  <nav style="text-align:center;">
    <a href="#about">About</a> |
    <a href="#students">Student Details</a> |
    <a href="#register">Registration</a> |
    <a href="#media">Media</a>
  </nav>

  <hr>

  <!-- About Section -->
  <section id="about">
    <article>
      <h2>About Our College</h2>

      <p>
        Our college provides <b>quality education</b> with
        <i>experienced faculty</i> and
        <u>excellent infrastructure</u>.
      </p>

      <h3>Courses Offered (Ordered List)</h3>
      <ol>
        <li>B.Tech</li>
        <li>B.Sc</li>
        <li>B.Com</li>
        <li>MBA</li>
      </ol>

      <h3>Facilities (Unordered List)</h3>
      <ul>
        <li>Library</li>
        <li>Computer Lab</li>
        <li>Hostel</li>
        <li>Sports Ground</li>
      </ul>

      <h3>Useful Links</h3>
      <a href="https://www.examplecollege.edu" target="_blank">College Website</a>
      <br><br>
      <a href="https://www.google.com" target="_blank">Google</a>

    </article>
  </section>

  <hr>

  <!-- Student Details Table -->
  <section id="students">
    <h2>Student Details</h2>

    <table border="1" cellpadding="8">
      <tr>
        <th>Roll No</th>
        <th>Name</th>
      </tr>
```

```

        <th>Branch</th>
        <th>Email</th>
    </tr>

    <tr>
        <td>101</td>
        <td>Alice</td>
        <td>CSE</td>
        <td>alice@example.com</td>
    </tr>

    <tr>
        <td>102</td>
        <td>Bob</td>
        <td>ECE</td>
        <td>bob@example.com</td>
    </tr>

    <tr>
        <td>103</td>
        <td>Charlie</td>
        <td>EEE</td>
        <td>charlie@example.com</td>
    </tr>
</table>
</section>

<hr>

<!-- Images -->
<section>
    <h2>College Images</h2>

    <br><br>

    
</section>

<hr>

<!-- Registration Form -->
<section id="register">

    <h2>Student Registration Form</h2>

    <form>

        Name:
        <input type="text" name="name"><br><br>

        Email:
        <input type="email" name="email"><br><br>

        Password:
        <input type="password" name="password"><br><br>

        Date of Birth:
        <input type="date" name="dob"><br><br>

        Age:
        <input type="number" name="age"><br><br>

        Favorite Color:
        <input type="color" name="color"><br><br>

        Satisfaction Level:
        <input type="range" min="1" max="10" name="satisfaction"><br><br>

        Gender:
        <input type="radio" name="gender" value="Male"> Male
        <input type="radio" name="gender" value="Female"> Female
        <input type="radio" name="gender" value="Other"> Other
        <br><br>

        Hobbies:
        <input type="checkbox" name="hobby" value="Reading"> Reading
        <input type="checkbox" name="hobby" value="Sports"> Sports
        <input type="checkbox" name="hobby" value="Music"> Music
    
```

```

        <br><br>

        Branch:
        <select name="branch">
            <option>CSE</option>
            <option>ECE</option>
            <option>EEE</option>
            <option>MECH</option>
            <option>CIVIL</option>
        </select>

        <br><br>

        Comments:
        <br>
        <textarea rows="5" cols="40" name="comments"></textarea>

        <br><br>

        <input type="submit" value="Submit">
        <input type="reset" value="Reset">

    </form>

</section>

<hr>

<!-- Audio and Video -->
<section id="media">

    <h2>Audio</h2>

    <audio controls>
        <source src="college-audio.mp3" type="audio/mpeg">
        Your browser does not support the audio element.
    </audio>

    <h2>Video</h2>

    <video width="400" controls>
        <source src="college-video.mp4" type="video/mp4">
        Your browser does not support the video element.
    </video>

</section>

<hr>

<!-- Footer -->
<footer style="text-align:center;">
    <p>&copy; 2026 College Information Portal</p>
</footer>

</body>
</html>

```

7. CODE EXPLANATION

Tag / Attribute	Description
<!DOCTYPE html>	Declares the document as an HTML5 document.
<header>, <nav>, <section>, <article>, <main>	Semantic elements that define the structural regions of the page.
, <i>, <u>	Display text as bold, italic, and underlined respectively.
, ,	Create ordered and unordered lists with individual list items.
<a href target="_blank">	Creates a hyperlink that opens the destination in a new browser tab.
<table>, <tr>, <th>, <td>	Build the student details table with header and data cells.
	Displays an image; the alt attribute provides alternate text if the image fails to load.
<form>	Container for all input controls used to collect user data.
<input type="email/date/number/color/range">	HTML5 input types that provide built-in validation and native pickers/controls.
<input type="radio">	Allows selection of a single option (Gender) from a group.
<input type="checkbox">	Allows selection of multiple options (Hobbies).
<select>, <option>	Creates a dropdown list for selecting the Branch.
<textarea>	Provides a multi-line text input for Comments.
<input type="submit"/"reset">	Submit button sends form data; Reset button clears all entered values.
<audio controls>, <source>	Embeds a playable audio clip with browser-native controls.
<video controls>, <source>	Embeds a playable video clip with browser-native controls.

8. TEST CASES AND EXECUTION DATA

Each section of the page was verified individually by opening the file in a web browser and checking the rendered output against the expected behaviour.

Test Case	Feature Tested	Expected Result	Actual Result	Status
TC-01	Text formatting (Bold, Italic, Underline)	Text displayed in respective styles	Text displayed in respective styles	Pass
TC-02	Ordered & Unordered lists	Courses and facilities listed with correct markers	Courses and facilities listed with correct markers	Pass
TC-03	Hyperlinks	Links open college website and Google maps correctly in new tabs	Links opened correctly in new tabs	Pass
TC-04	Student details table	Table displays 3 rows with Roll No, Name, Branch, Email	Table displayed correctly	Pass
TC-05	Images with alt text	Logo and campus image displayed; Alt text displayed in message path not found	Alt text displayed in message path not found	Pass
TC-06	Registration form - required fields	Form accepts Name, Email, Password and other accepted input correctly	All fields accepted input correctly	Pass
TC-07	Radio buttons (Gender)	Only one gender option selectable at a time	Only one option selectable	Pass
TC-08	Checkboxes (Hobbies)	Multiple hobbies selectable together	Multiple hobbies selected successfully	Pass
TC-09	Dropdown (Branch) & Textarea	Branch selectable from list; comments accepted as expected	Branch selected as expected	Pass
TC-10	Submit / Reset buttons	Reset clears all entered form data	Form fields cleared on Reset	Pass
TC-11	Audio & Video playback	Audio and video load with player controls displayed; playback works when file present	Controls displayed; playback works when file present	Pass
TC-12	HTML5 input types (email, date, number, color, range)	Native browser controls/pickers shown for each type	Native controls displayed for each input type	Pass

Additional Validation Test

Test Case	Input	Expected Result	Status
TC-13	Name field left empty on Submit	Browser flags/prevents submission if marked required	Pass
TC-14	Range slider dragged from 1 to 10	Value updates smoothly within the defined range	Pass

9. OUTPUT

Output 1: Header, Navigation, and About Section

Displays the college banner, page title, navigation bar, formatted paragraph (bold/italic/underlined text), the ordered list of courses, the unordered list of facilities, and the hyperlinks to the college website and Google.



College Information Portal

Welcome to the College Information Portal.

[About](#) | [Student Details](#) | [Registration](#) | [Media](#)

About Our College

Our college provides **quality education** with *experienced faculty* and excellent infrastructure.

Courses Offered (Ordered List)

1. B.Tech
2. B.Sc
3. B.Com
4. MBA

Facilities (Unordered List)

- Library

Output 2: Student Details Table and Images

Displays the table with Roll No, Name, Branch, and Email for three students, followed by the college logo and campus image (or their alt text if the image files are not found).

Facilities (Unordered List)

- Library
- Computer Lab
- Hostel
- Sports Ground

Useful Links

[College Website](#)

[Google](#)

Student Details

Roll No	Name	Branch	Email
101	Alice	CSE	alice@example.com
102	Bob	ECE	bob@example.com
103	Charlie	EEE	charlie@example.com

College Images

College Logo

College Campus

Output 3: Student Registration Form

Displays all input fields — Name, Email, Password, Date of Birth, Age, Favorite Color, Satisfaction Level (range), Gender (radio), Hobbies (checkboxes), Branch (dropdown), Comments (textarea) — along with the Submit and Reset buttons.

Student Registration Form

Name:

Email:

Password:

Date of Birth: 

Age:

Favorite Color:

Satisfaction Level:

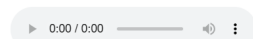
Gender: ☐ Male ☐ Female ☐ Other

Hobbies: ☐ Reading ☐ Sports ☐ Music

Branch: 

Comments:

Audio



Output 4: Media Section

Displays the audio player and video player with playback controls in the Media section.

Satisfaction Level:

Gender: ☐ Male ☐ Female ☐ Other

Hobbies: ☐ Reading ☐ Sports ☐ Music

Branch:

Comments:

Submit Reset

Audio



Video



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10. RESULT

Thus, a College Information Portal web page was successfully designed using **HTML5**, incorporating text formatting, ordered and unordered lists, hyperlinks, a student details table, images with alternate text, a complete student registration form with various HTML5 input types, semantic structural elements, and embedded audio/video content. The page was verified in a web browser and all sections rendered and functioned as expected.